

Maulana Abul Kalam Azad University of Technology, West Bengal
(Formerly known as West Bengal University of Technology)
Syllabus of *B.Sc. in Data Science*
Effective from academic session 2023-2024

Paper Code: BDS402 & BDS492

Paper Name: *Artificial Intelligence & Machine Learning - 2 & AI ML-2 Lab*

Credits: 4 (2Th+2P)

1. Introduction to Artificial Neural Networks

- Historical overview and motivations
- Biological inspiration and neuron model
- Basic components of neural networks
- Types of neural networks (e.g. feedforward, recurrent)

2. Perceptrons and Activation Functions

- McCulloch - Pitt Neuron & Single-layer perceptron model
- Activation functions and their properties
- Multilayer perceptron architecture
- Training and learning algorithms for perceptrons
- Appropriate laboratory work in Python/R

3. Feedforward Neural Networks

- Feedforward network structure and architecture
- Forward propagation and activation functions
- Backpropagation algorithm for training
- Overfitting, underfitting, and regularization techniques
- Appropriate laboratory work in Python/R

4. Radial Basis Function Networks

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- Introduction to radial basis functions
- RBF network architecture and training
- Applications of RBF networks in pattern recognition
- Appropriate laboratory work in Python/R
- 5. Self-Organizing Maps**
- Introduction to Kohonen Self-Organizing Maps (SOMs)
- SOM architecture and learning algorithm
- Clustering and visualization using SOMs
- Applications of SOMs in data analysis
- Appropriate laboratory work in Python/R

6. Hopfield Networks

- Introduction to associative memory
- Hopfield network architecture and energy function
- Pattern storage and retrieval in Hopfield networks
- Applications of Hopfield networks in optimization problems
- Appropriate laboratory work in Python/R

7. Neural Networks in Pattern Recognition

- Feature extraction and selection techniques
- Handwritten digit recognition using neural networks
- Face recognition using neural networks
- Applications of neural networks in image classification
- Appropriate laboratory work in Python/R

References:

1. "Neural Networks and Learning Machines" by Simon Haykin. Publisher: Pearson Date of Publication: 2021
2. "Deep Learning & Neural Networks" by M.C. Trivedi, Khanna Publishing House: 2024.
3. "Pattern Recognition and Machine Learning" by Christopher Bishop Publisher: Springer, 2006
4. "Neural Networks and Deep Learning" by Charu C. Aggarwal, Springer
5. "Principles of Soft Computing. 2nd Edition" by Sivanandam, S.N. and Deepa, S.N. (2009), John Wiley & Sons, Inc.